

7

Ratio and Proportion

(Including Properties and Uses)

7.1 Introduction :

Basic concepts of ratio and proportion have already been studied in earlier classes, especially in classes 8 and 9. In this chapter of Class 10, we shall study ratio and proportion in more detail.

7.2 Ratio :

The ratio of two quantities of the same kind and in the same units is a comparison obtained by dividing the first quantity by the other.

If a and b are two quantities of the same kind and with the same units such that $b \neq 0$; then the quotient $\frac{a}{b}$ is called the **ratio** between a and b .

Remember :

1. Ratio $\frac{a}{b}$ has no unit and can be written as $a : b$ (read as a is to b).
2. The quantities a and b are called terms of the ratio. The first quantity a is called the *first term* or the *antecedent* and the second quantity b is called the *second term* or the *consequent* of the ratio $a : b$.

The second term of a ratio cannot be zero.

- i.e.* (i) In the ratio $a : b$, the second term b cannot be zero ($b \neq 0$).
- (ii) In the ratio $b : a$, the second term $a \neq 0$.
3. If both the terms of a ratio are multiplied or divided by the same non-zero number, the ratio remains unchanged.
 4. A ratio must always be expressed in its *lowest terms* *i.e.* both the terms of the ratio must be co-prime.

The ratio is in its lowest terms, if the H.C.F. of its both the terms is 1 (unity).

- e.g.* (i) The ratio $3 : 7$ is in its lowest terms as the H.C.F. of its terms 3 and 7 is 1.
- (ii) The ratio $4 : 20$ is not in its lowest terms as the H.C.F. of its terms 4 and 20 is 4 and not 1.
5. Ratios $a : b$ and $b : a$ cannot be equal unless $a = b$
i.e., $a : b \neq b : a$, unless $a = b$.

In other words, the order of the terms in a ratio is important.

- 1 (i) If $2x + 3y : 3x + 5y = 18 : 29$, find $x : y$.
- (ii) If $x : y = 2 : 3$, find the value of $3x + 2y : 2x + 5y$.

Solution :

$$(i) \quad 2x + 3y : 3x + 5y = 18 : 29$$

$$\Rightarrow \frac{2x + 3y}{3x + 5y} = \frac{18}{29}$$

$$\Rightarrow 58x + 87y = 54x + 90y$$

$$\Rightarrow 4x = 3y$$

$$\Rightarrow \frac{x}{y} = \frac{3}{4} \quad \text{i.e. } x : y = 3 : 4$$

Ans.

$$(ii) \quad x : y = 2 : 3 \Rightarrow \frac{x}{y} = \frac{2}{3}$$

$$\text{Now, } 3x + 2y : 2x + 5y = \frac{3x + 2y}{2x + 5y}$$

$$= \frac{3\left(\frac{x}{y}\right) + 2}{2\left(\frac{x}{y}\right) + 5}$$

[Dividing each term by y]

$$= \frac{3 \times \frac{2}{3} + 2}{2 \times \frac{2}{3} + 5}$$

$$[\because \frac{x}{y} = \frac{2}{3}]$$

$$= 12 : 19$$

Ans.**Alternative method :**

$$\Rightarrow x : y = 2 : 3 \Rightarrow 3x = 2y \Rightarrow x = \frac{2y}{3}$$

$$\therefore \frac{3x + 2y}{2x + 5y} = \frac{3 \times \frac{2y}{3} + 2y}{2 \times \frac{2y}{3} + 5y} = \frac{4y}{\frac{19y}{3}} = \frac{4 \times 3}{19} = 12 : 19$$

Ans.

$$\text{OR, } x : y = 2 : 3 \Rightarrow 3x = 2y \Rightarrow y = \frac{3x}{2}$$

$$\therefore \frac{3x + 2y}{2x + 5y} = \frac{3x + 2 \times \frac{3x}{2}}{2x + 5 \times \frac{3x}{2}} = \frac{6x}{\frac{19x}{2}} = \frac{6 \times 2}{19} = 12 : 19$$

Ans.**3rd Method :**

$$x : y = 2 : 3 \Rightarrow \text{if } x = 2k \text{ then } y = 3k$$

$$\text{And, } \frac{3x + 2y}{2x + 5y} = \frac{3 \times 2k + 2 \times 3k}{2 \times 2k + 5 \times 3k} = \frac{12k}{19k} = \frac{12}{19} = 12 : 19$$

Ans.

Precaution :

For $x : y = 2 : 3$; if we take $x = 2$ and $y = 3$; then

$$\frac{3x + 2y}{2x + 5y} = \frac{3 \times 2 + 2 \times 3}{2 \times 2 + 5 \times 3}$$

$$= \frac{12}{19} = 12 : 19; \text{ which is the same as obtained in each solution given above.}$$

But this solution is absolutely wrong and for this solution, a student will score no marks.

Reason : Let the age of Mohit = 15 yrs. and the age of his elder brother Rahul = 24 yrs. The ratio between the ages of Mohit and Rahul = 15 yrs : 24 yrs = 5 : 8. Now read it otherwise, that the ratio between the ages of Mohit and Rahul is 5 : 8. What does it mean ? Does it mean that Mohit's age is 5 years and Rahul's age is 8 years. The answer is simple, i.e. No.

In the same way, if $x : y = 2 : 3$, it does not mean $x = 2$ and $y = 3$.

2 If $a : b = 5 : 3$, find $(5a + 8b) : (6a - 7b)$.

[2002]

Solution :

Let $a : b = 5 : 3 \Rightarrow$ if $a = 5x$, then $b = 3x$;

$$\text{and } \frac{5a + 8b}{6a - 7b} = \frac{5 \times 5x + 8 \times 3x}{6 \times 5x - 7 \times 3x} = \frac{49x}{9x} = 49 : 9 \quad \text{Ans.}$$

3 Two numbers are in the ratio 3 : 5. If 8 is added to each number, the ratio becomes 2 : 3. Find the numbers.

Solution :

Since, the ratio between the numbers is 3 : 5

$\Rightarrow$ if one number is $3x$; the other number is $5x$

$$\text{Given : } \frac{3x + 8}{5x + 8} = \frac{2}{3} \quad \Rightarrow \quad 10x + 16 = 9x + 24$$

$$\Rightarrow \quad x = 8$$

$\therefore$ Nos. are $3x$ and $5x = 3 \times 8$ and $5 \times 8 = 24$ and 40

Ans.

4 (i) What quantity must be added to each term of the ratio 8 : 15 so that it becomes equal to 3 : 5 ?

(ii) What quantity must be subtracted from each term of the ratio $a : b$ so that it becomes $c : d$?

Solution :

(i) Let x be added to each term of the ratio 8 : 15.

$$\text{Given : } \frac{8 + x}{15 + x} = \frac{3}{5}$$

$$\Rightarrow \quad 40 + 5x = 45 + 3x \quad \Rightarrow \quad x = 2\frac{1}{2}$$

Ans.

(ii) Let x be subtracted, then :

$$\frac{a-x}{b-x} = \frac{c}{d}$$

$$\Rightarrow ad - dx = bc - cx$$

$$\Rightarrow cx - dx = bc - ad$$

$$\Rightarrow x(c - d) = bc - ad \Rightarrow x = \frac{bc - ad}{c - d}$$

Ans.

- 5** The work done by $(x - 3)$ men in $(2x + 1)$ days and the work done by $(2x + 1)$ men in $(x + 4)$ days are in the ratio 3 : 10. Find the value of x . [2003]

Solution :

Assuming that all the men do the same amount of work in one day and one day work of each man = 1 unit; we get :

Amount of work done by $(x - 3)$ men in $(2x + 1)$ days

= amount of work done by $(x - 3)$ $(2x + 1)$ men in one day

= $(x - 3)(2x + 1)$ units of work.

Similarly, amount of work done by $(2x + 1)$ men in $(x + 4)$ days.

= amount of work done by $(2x + 1)(x + 4)$ men in one day.

= $(2x + 1)(x + 4)$ units of work.

According to the given statement :

$$\frac{(x-3)(2x+1)}{(2x+1)(x+4)} = \frac{3}{10}$$

$$\Rightarrow \frac{2x^2 + x - 6x - 3}{2x^2 + 8x + x + 4} = \frac{3}{10} \quad \text{i.e.} \quad \frac{2x^2 - 5x - 3}{2x^2 + 9x + 4} = \frac{3}{10}$$

$$\Rightarrow 20x^2 - 50x - 30 = 6x^2 + 27x + 12$$

$$\Rightarrow 14x^2 - 77x - 42 = 0$$

$$\Rightarrow 2x^2 - 11x - 6 = 0$$

$$\Rightarrow (x - 6)(2x + 1) = 0$$

[On factorising]

$$\Rightarrow x = 6, \text{ or } x = -\frac{1}{2}$$

$x = -\frac{1}{2}$ is not possible as it will make no. of men $(x - 3)$ negative.

$$\therefore x = 6$$

Ans.

7.3 Increase (or decrease) in a ratio :

1. Let the price of an article increases from ₹ 20 to ₹ 24; we say that the price has increased in the ratio $20 : 24 = 5 : 6$.

$\Rightarrow$ The original price of the article : Its increased price = 5 : 6

2. Let the price of an article decreases from ₹ 24 to ₹ 20; we say that the price has decreased in the ratio $24 : 20 = 6 : 5$.

$\Rightarrow$ The original price of the article : Its decreased price = 6 : 5

In general :

If a quantity increases or decreases in the ratio $a : b$.

$\Rightarrow$ The new (resulting) quantity = $\frac{b}{a}$ times of the original quantity.

- 6** When the fare of a certain journey by an airliner was increased in the ratio 5 : 7 the cost of the ticket for the journey became ₹ 1,421. Find the increase in the fare.

Solution :

According to the given statement :

The original fair : Increased fair = 5 : 7

$$\Rightarrow 7 \times \text{The original fare} = 5 \times \text{Increased fair}$$

$$\Rightarrow 7 \times \text{The original fare} = 5 \times ₹ 1,421$$

$$\Rightarrow \text{The original fare} = \frac{5 \times ₹ 1,421}{7} = ₹ 1,015$$

$$\therefore \text{Increase in the fare} = ₹ (1,421 - 1,015) = ₹ 406 \quad \text{Ans.}$$

- 7** In a regiment, the ratio of number of officers to the number of soldiers was 3 : 31 before a battle. In the battle 6 officers and 22 soldiers were killed. The ratio between the number of officers and the number of soldiers now is 1 : 13. Find the number of officers and soldiers in the regiment before the battle. [1992]

Solution :**Before the battle :**

Let the number of officers be $3x$

$$\Rightarrow \text{the number of soldiers} = 31x$$

After the battle :

The number of officers = $3x - 6$

and, the number of soldiers = $31x - 22$

$$\text{Given : } \frac{3x-6}{31x-22} = \frac{1}{13} \Rightarrow x = 7 \quad [\text{On solving}]$$

$$\therefore \text{The no. of officers before battle} = 3x = 3 \times 7 = 21$$

$$\text{and the no. of soldiers before battle} = 31x = 31 \times 7 = 217 \quad \text{Ans.}$$

- 8** If $\frac{a}{b+c} = \frac{b}{c+a} = \frac{c}{a+b}$ and $a + b + c = 0$; show that each given ratio is equal to -1 .

Solution :

$$\begin{aligned} \text{Since, } a + b + c = 0 &\Rightarrow a + b = -c, \\ &b + c = -a \text{ and } c + a = -b \end{aligned}$$

$$\therefore \frac{a}{b+c} = \frac{a}{-a} = -1; \frac{b}{c+a} = \frac{b}{-b} = -1 \text{ and } \frac{c}{a+b} = \frac{c}{-c} = -1$$

Hence, **each of the given ratios is -1.**

Ans.

9 If $\frac{a}{b+c} = \frac{b}{c+a} = \frac{c}{a+b}$ and $a + b + c \neq 0$; show that each given ratio is equal to $\frac{1}{2}$.

Solution :

For any two or more equal ratios, each ratio is equal to the ratio between sum of their antecedents and sum of their consequents.

$$\therefore \text{ (i) } \frac{a}{b} = \frac{c}{d} \Rightarrow \frac{a}{b} = \frac{c}{d} = \frac{a+c}{b+d}$$

$$\text{ (ii) } \frac{a}{b} = \frac{c}{d} = \frac{e}{f} \Rightarrow \frac{a}{b} = \frac{c}{d} = \frac{e}{f} = \frac{a+c+e}{b+d+f} \text{ and so on.}$$

$$\text{Given : } \frac{a}{b+c} = \frac{b}{c+a} = \frac{c}{a+b}$$

$$\begin{aligned} \Rightarrow \frac{a}{b+c} &= \frac{b}{c+a} = \frac{c}{a+b} = \frac{\text{sum of antecedents}}{\text{sum of consequents}} \\ &= \frac{a+b+c}{(b+c)+(c+a)+(a+b)} \\ &= \frac{a+b+c}{2a+2b+2c} = \frac{a+b+c}{2(a+b+c)} \\ &= \frac{1}{2} \end{aligned}$$

Ans

7.4 Commensurable and incommensurable quantities :

If the ratio between any two quantities of the same kind and having the same unit can be expressed exactly by the ratio between two integers; the quantities are said to be *commensurable*; otherwise *incommensurable*,

e.g. (i) The ratio between $2\frac{1}{3}$ and $3\frac{1}{2}$

$$= \frac{7}{3} : \frac{7}{2} = \frac{7}{3} \times \frac{2}{7} = 2 : 3; \text{ which is the ratio between two integers 2 and 3.}$$

Therefore, $2\frac{1}{3}$ and $3\frac{1}{2}$ are *commensurable quantities*.

(ii) The ratio between $\sqrt{3}$ and 5 is $\sqrt{3} : 5$; which can never be expressed as the ratio between two integers; therefore $\sqrt{3}$ and 5 are *incommensurable quantities*.

7.5 Composition of Ratios :**(i) Compound Ratio :**

For two or more ratios, the ratio between the product of their antecedents to the product of their consequents is called **compound ratio**.

e.g., For ratios $a : b$ and $c : d$; the compound ratio is $(a \times c) : (b \times d)$.

For ratios $a : b$, $c : d$ and $e : f$; the compound ratio is

$$(a \times c \times e) : (b \times d \times f) \text{ and so on.}$$

(ii) Duplicate Ratio :

It is the compound ratio of two equal ratios.

e.g., **Duplicate ratio of $a : b$** = Compound ratio of $a : b$ and $a : b$
 $= (a \times a) : (b \times b) = a^2 : b^2$

$$\begin{aligned} \text{Thus, duplicate ratio of } 2 : 3 &= 2^2 : 3^2 \\ &= 4 : 9 \end{aligned}$$

(iii) Triplicate Ratio :

It is the compound ratio of three equal ratios.

e.g., **Triplicate ratio of $a : b$** = Compound ratio of $a : b$, $a : b$ and $a : b$
 $= (a \times a \times a) : (b \times b \times b) = a^3 : b^3$

$$\begin{aligned} \text{Thus, triplicate ratio of } 2 : 3 &= 2^3 : 3^3 \\ &= 8 : 27 \end{aligned}$$

(iv) Sub-duplicate Ratio :

For any ratio $a : b$, its sub-duplicate ratio is $\sqrt{a} : \sqrt{b}$

$$\begin{aligned} \text{Thus, sub-duplicate ratio of } 9 : 16 &= \sqrt{9} : \sqrt{16} \\ &= 3 : 4. \end{aligned}$$

(v) Sub-triplicate Ratio :

For any ratio $a : b$, its sub-triplicate ratio is $\sqrt[3]{a} : \sqrt[3]{b}$

$$\begin{aligned} \text{Thus, sub-triplicate ratio of } 27 : 64 &= \sqrt[3]{27} : \sqrt[3]{64} \\ &= 3 : 4 \end{aligned}$$

(vi) Reciprocal Ratio :

For any ratio $a : b$; where $a, b \neq 0$, its reciprocal ratio is $\frac{1}{a} : \frac{1}{b} = b : a$.

$$\text{Thus, reciprocal ratio of } 3 : 5 = \frac{1}{3} : \frac{1}{5} = 5 : 3.$$

10 Find the compound ratio of :

(i) $3a : 2b$, $2m : n$ and $4x : 3y$

(ii) $a - b : a + b$, $(a + b)^2 : a^2 + b^2$ and $a^4 - b^4 : (a^2 - b^2)^2$.

Solution :

(i) **Required compound ratio** $= (3a \times 2m \times 4x) : (2b \times n \times 3y)$

$$= \frac{24 amx}{6 bny} = 4 amx : bny$$

Ans.

(ii) **Required compound ratio** $= [(a-b) \cdot (a+b)^2 \cdot (a^4-b^4)] : [(a+b) \cdot (a^2+b^2) \cdot (a^2-b^2)^2]$

$$= \frac{(a-b)(a+b)^2(a^2+b^2)(a^2-b^2)}{(a+b)(a^2+b^2)(a^2-b^2)(a+b)(a-b)}$$

$$= 1 : 1$$

Ans.

- 11** Find the ratio compounded of the duplicate ratio of 5 : 6, the reciprocal ratio of 25 : 42 and the sub-triplicate ratio of 216 : 343.

Solution :

Since, duplicate ratio of 5 : 6 $= 5^2 : 6^2 = 25 : 36$,

reciprocal ratio of 25 : 42 $= \frac{1}{25} : \frac{1}{42} = 42 : 25$

and, sub-triplicate ratio of 216 : 343 $= \sqrt[3]{216} : \sqrt[3]{343} = 6 : 7$.

Therefore, **the required compounded ratio** $= (25 \times 42 \times 6) : (36 \times 25 \times 7)$

$$= \frac{25 \times 42 \times 6}{36 \times 25 \times 7} = 1 : 1$$

Ans.

EXERCISE 7(A)

- If $a : b = 5 : 3$, find : $\frac{5a-3b}{5a+3b}$.
- If $x : y = 4 : 7$, find the value of $(3x+2y) : (5x+y)$.
- If $a : b = 3 : 8$, find the value of $\frac{4a+3b}{6a-b}$.
- If $(a-b) : (a+b) = 1 : 11$, find the ratio $(5a+4b+15) : (5a-4b+3)$.
- Find the number which bears the same ratio to $\frac{7}{33}$ that $\frac{8}{21}$ does to $\frac{4}{9}$.
- If $\frac{m+n}{m+3n} = \frac{2}{3}$, find : $\frac{2n^2}{3m^2+mn}$.
- Find $\frac{x}{y}$; when $x^2 + 6y^2 = 5xy$.
- If the ratio between 8 and 11 is the same as the ratio of $2x-y$ to $x+2y$, find the value of $\frac{7x}{9y}$.
- Divide ₹ 1,290 into A, B and C such that A is $\frac{2}{5}$ of B and $B : C = 4 : 3$.
- A school has 630 students. The ratio of the number of boys to the number of girls is 3 : 2. This ratio changes to 7 : 5 after the admission of 90 new students. Find the number of newly admitted boys.
- What quantity must be subtracted from each term of the ratio 9 : 17 to make it equal to 1 : 3?
- The monthly pocket money of Ravi and Sanjeev are in the ratio 5 : 7. Their expenditures are in the ratio 3 : 5. If each saves ₹ 80 every month, find their monthly pocket money. **[2012]**
- The work done by $(x-2)$ men in $(4x+1)$ days and the work done by $(4x+1)$ men in $(2x-3)$ days are in the ratio 3 : 8. Find the value of x .
- The bus fare between two cities is increased in the ratio 7 : 9. Find the increase in the fare, if :
 - the original fare is ₹ 245;
 - the increased fare is ₹ 207.
- By increasing the cost of entry ticket to a fair in the ratio 10 : 13, the number of visitors to the fair has decreased in the ratio 6 : 5. In

what ratio has the total collection increased or decreased ?

16. In a basket, the ratio between the number of oranges and the number of apples is 7 : 13. If 8 oranges and 11 apples are eaten, the ratio between the number of oranges and the number of apples becomes 1 : 2. Find the original number of oranges and the original number of apples in the basket.
17. In a mixture of 126 kg of milk and water, milk and water are in ratio 5 : 2. How much water must be added to the mixture to make this ratio 3 : 2 ?
18. (a) If $A : B = 3 : 4$ and $B : C = 6 : 7$, find :
(i) $A : B : C$
(ii) $A : C$
(b) If $A : B = 2 : 5$ and $A : C = 3 : 4$, find :
 $A : B : C$.
19. (i) If $3A = 4B = 6C$; find : $A : B : C$.
(ii) If $2a = 3b$ and $4b = 5c$, find : $a : c$.
20. Find the compound ratio of :
(i) 2 : 3, 9 : 14 and 14 : 27.
(ii) $2a : 3b$, $mn : x^2$ and $x : n$.
(iii) $\sqrt{2} : 1$, $3 : \sqrt{5}$ and $\sqrt{20} : 9$.

21. Find duplicate ratio of :

- (i) 3 : 4 (ii) $3\sqrt{3} : 2\sqrt{5}$

22. Find triplicate ratio of :

- (i) 1 : 3 (ii) $\frac{m}{2} : \frac{n}{3}$

23. Find sub-duplicate ratio of :

- (i) 9 : 16 (ii) $(x - y)^4 : (x + y)^6$

24. Find sub-triplicate ratio of :

- (i) 64 : 27 (ii) $x^3 : 125y^3$

25. Find the reciprocal ratio of :

- (i) 5 : 8 (ii) $\frac{x}{3} : \frac{y}{7}$

26. If $(x + 3) : (4x + 1)$ is the duplicate ratio of 3 : 5, find the value of x .

27. If $m : n$ is the duplicate ratio of $m + x : n + x$; show that : $x^2 = mn$.

28. If $(3x - 9) : (5x + 4)$ is the triplicate ratio of 3 : 4, find the value of x .

29. Find the ratio compounded of the reciprocal ratio of 15 : 28, the sub-duplicate ratio of 36 : 49 and the triplicate ratio of 5 : 4.

30. (a) If $r^2 = pq$, show that $p : q$ is the duplicate ratio of $(p + r) : (q + r)$.

(b) If $(p - x) : (q - x)$ be the duplicate ratio of $p : q$ then show that : $\frac{1}{p} + \frac{1}{q} = \frac{1}{x}$.

7.6 Proportion :

Four non-zero quantities, a , b , c and d are said to be in proportion (or, are proportional), if $a : b = c : d$.

This is often expressed as $a : b :: c : d$ and is read as “ a is to b as c is to d ”.

1. In $a : b = c : d$,

(i) a , b , c and d are called the terms of the proportion; where a = first term, b = second term, c = third term and d = fourth term.

(ii) ‘ a ’ and ‘ d ’ are called *extremes* (end-terms) whereas ‘ b ’ and ‘ c ’ are called *means* (middle terms).

$$2. a : b = c : d \Rightarrow \frac{a}{b} = \frac{c}{d} \Rightarrow a \times d = b \times c$$

$\Rightarrow$ **product of extremes = product of means.**

3. In $a : b = c : d$, the fourth term ‘ d ’ is called the *fourth proportional*.

4. In $a : b = c : d$, quantities a and b must be of the same kind with the same units, whereas; c and d may separately be of the same kind with the same units.

e.g. 5kg : 15 kg = ₹ 75 : ₹ 225

7.7 Continued proportion :

Three non-zero quantities of the same kind and in the same unit are said to be in **continued proportion**, if the ratio of the first to the second is the same as the ratio of the second to the third.

Thus, a , b and c are in the continued proportion if $a : b = b : c$.

In general, the non-zero quantities a , b , c , d , e , etc. (all of the same kind and in the same unit) are in continued proportion $\Leftrightarrow \frac{a}{b} = \frac{b}{c} = \frac{c}{d} = \frac{d}{e}$

a , b and c are in continued proportion

$$\Leftrightarrow a : b = b : c,$$

Here, the *second quantity* i.e. ' b ' is called the *mean proportional* between ' a ' and ' c '; whereas the *third quantity* i.e. ' c ' is called the *third proportional* to ' a ' and ' b '.

- 12** Find : (i) the fourth proportional to 3, 6 and 4.5.
 (ii) the mean proportional between 6.25 and 0.16.
 (iii) the third proportional to 1.2 and 1.8.

Solution :

- (i) Let the fourth proportional to 3, 6 and 4.5 be x .

$$\Rightarrow 3 : 6 = 4.5 : x$$

$$\Rightarrow 3 \times x = 6 \times 4.5 \quad \Rightarrow \quad x = 9$$

Ans.

- (ii) Let the mean proportional between 6.25 and 0.16 be x .

$$\Rightarrow 6.25, x \text{ and } 0.16 \text{ are in continued proportion.}$$

$$\Rightarrow 6.25 : x = x : 0.16$$

$$\Rightarrow x \times x = 6.25 \times 0.16 \quad \Rightarrow \quad x^2 = 1 \quad \Rightarrow \quad x = 1$$

Ans.

- (iii) Let the third proportional to 1.2 and 1.8 be x

$$\Rightarrow 1.2, 1.8 \text{ and } x \text{ are in continued proportion.}$$

$$\Rightarrow 1.2 : 1.8 = 1.8 : x \quad \Rightarrow \quad x = \frac{1.8 \times 1.8}{1.2} = 2.7$$

Ans.

- 13** Quantities a , 2, 10 and b are in continued proportion; find the values of a and b .

Solution :

a , 2, 10 and b are in continued proportion

$$\Rightarrow \frac{a}{2} = \frac{2}{10} = \frac{10}{b} \quad \Rightarrow \quad \frac{a}{2} = \frac{2}{10} \text{ and } \frac{2}{10} = \frac{10}{b}$$

$$\Rightarrow a = 0.4 \text{ and } b = 50$$

Ans.

- 14** What number should be subtracted from each of the numbers 23, 30, 57 and 78; so that the remainders are in proportion ? [2004]

Solution :

Let the number subtracted be x .

$$\therefore (23 - x) : (30 - x) :: (57 - x) : (78 - x)$$

$$\Rightarrow \frac{23-x}{30-x} = \frac{57-x}{78-x}$$

$$\Rightarrow 1794 - 101x + x^2 = 1710 - 87x + x^2 \Rightarrow 14x = 84 \text{ and } x = 6 \quad \text{Ans.}$$

- 15** What should be added to each of the numbers 13, 17 and 22 so that the resulting numbers are in continued proportion ?

Solution :

Let the required number to be added is x .

$\therefore 13 + x, 17 + x$ and $22 + x$ are in continued proportion.

$$\Rightarrow \frac{13+x}{17+x} = \frac{17+x}{22+x} \quad \text{i.e. } (13+x)(22+x) = (17+x)^2$$

$$\Rightarrow 286 + 35x + x^2 = 289 + 34x + x^2 \quad \text{i.e. } x = 3$$

$\therefore$ **Required number = 3** **Ans.**

- 16** If $(a^2 + c^2)$, $(ab + cd)$ and $(b^2 + d^2)$ are in continued proportion; prove that a , b , c and d are in proportion.

Solution :

Given, $(a^2 + c^2)$, $(ab + cd)$ and $(b^2 + d^2)$ are in continued proportion.

$$\therefore \frac{a^2+c^2}{ab+cd} = \frac{ab+cd}{b^2+d^2} \quad \Rightarrow (a^2 + c^2)(b^2 + d^2) = (ab + cd)^2$$

$$\text{i.e. } a^2b^2 + a^2d^2 + b^2c^2 + c^2d^2 = a^2b^2 + 2abcd + c^2d^2$$

$$\Rightarrow a^2d^2 + b^2c^2 - 2abcd = 0 \quad \text{i.e. } (ad - bc)^2 = 0$$

$$\Rightarrow ad - bc = 0 \quad \text{i.e. } ad = bc$$

$$\Rightarrow \frac{a}{b} = \frac{c}{d} \quad \text{i.e. } a, b, c \text{ and } d \text{ are in proportion}$$

Hence Proved.

- 17** If $p : q :: q : r$, prove that $p : r = p^2 : q^2$.

Solution :

$$p : q :: q : r \Rightarrow q^2 = pr$$

$$\begin{aligned}\therefore p^2 : q^2 &= \frac{p^2}{q^2} = \frac{p^2}{pr} & [\because q^2 = pr] \\ &= \frac{p}{r} = p : r. & \text{Hence Proved.}\end{aligned}$$

- 18** If $a \neq b$ and $a : b$ is the duplicate ratio of $a + c$ and $b + c$, prove that 'c' is the mean proportional between 'a' and 'b'.

Solution :

'c' will be mean proportional between 'a' and 'b', if $a : c = c : b$ i.e., if $c^2 = ab$.

$$\begin{aligned}\text{Given :} \quad \frac{a}{b} &= \frac{(a+c)^2}{(b+c)^2} \\ \Rightarrow a(b^2 + c^2 + 2bc) &= b(a^2 + c^2 + 2ac) \\ \Rightarrow ab^2 + ac^2 + 2abc &= a^2b + bc^2 + 2abc \\ \Rightarrow ac^2 - bc^2 &= a^2b - ab^2 \\ \Rightarrow c^2(a - b) &= ab(a - b) \\ \Rightarrow c^2 &= ab & [\text{As } a \neq b] \\ \Rightarrow \text{'c' is mean proportional between 'a' and 'b'}. & \text{Hence Proved.}\end{aligned}$$

- 19** If $a + c = mb$ and $\frac{1}{b} + \frac{1}{d} = \frac{m}{c}$, prove that a, b, c and d are in proportion.

Solution :

$$\begin{aligned}\frac{1}{b} + \frac{1}{d} &= \frac{m}{c} & \Rightarrow \quad \frac{d+b}{bd} = \frac{m}{c} \\ \text{i.e.} \quad cd + cb &= mbd \\ \Rightarrow cd + bc &= (a+c)d & [\because a+c=mb] \\ \Rightarrow cd + bc &= ad + cd \\ \Rightarrow bc &= ad & \text{i.e.} \quad \frac{a}{b} = \frac{c}{d} \\ \Rightarrow \text{a, b, c and d are in proportion} & \text{Hence Proved.}\end{aligned}$$

Alternative method :

$$\begin{aligned}a + c &= mb & \Rightarrow \quad m = \frac{a+c}{b} \\ \text{Substituting the value of } m &\text{ in the other given equation, we get :} \\ \frac{1}{b} + \frac{1}{d} &= \frac{a+c}{bc} & \Rightarrow \quad \frac{d+b}{bd} = \frac{a+c}{bc} \\ \text{i.e.} \quad \frac{d+b}{d} &= \frac{a+c}{c} & \Rightarrow \quad cd + bc = ad + cd\end{aligned}$$

i.e.

$$bc = ad$$

 $\Rightarrow$

$$\frac{a}{b} = \frac{c}{d}$$

i.e. a, b, c and d are in proportion

Hence Proved.

20 If q is the mean proportional between p and r , prove that :

$$p^2 - q^2 + r^2 = q^4 \left(\frac{1}{p^2} - \frac{1}{q^2} + \frac{1}{r^2} \right).$$

Solution : $\therefore q$ is the mean proportional between p and $r \Rightarrow q^2 = pr$

$$\therefore \text{R.H.S.} = q^4 \left(\frac{1}{p^2} - \frac{1}{q^2} + \frac{1}{r^2} \right)$$

$$= \frac{q^4}{p^2} - q^2 + \frac{q^4}{r^2}$$

$$= \frac{p^2 r^2}{p^2} - q^2 + \frac{p^2 r^2}{r^2}$$

$$[q^2 = pr \Rightarrow q^4 = p^2 r^2]$$

$$= r^2 - q^2 + p^2 = \text{L.H.S.}$$

Hence Proved.

Alternative method ('k' method) :

- Step :**
1. Put each given ratios equal to k .
 2. Obtain the antecedent of each ratio in terms of k .
 3. Substitute the values, obtained in step 2 in terms of k .
 4. Simplify.

Given : q is the mean proportional between p and r

$$\Rightarrow p : q = q : r$$

$$\Rightarrow \frac{p}{q} = \frac{q}{r} = k \text{ (say)} \quad \Rightarrow \frac{p}{q} = k \text{ and } \frac{q}{r} = k$$

$$\text{i.e. } p = qk, q = rk \text{ and } p = qk = (rk)k = rk^2$$

$$\therefore \text{R.H.S.} = q^4 \left(\frac{1}{p^2} - \frac{1}{q^2} + \frac{1}{r^2} \right)$$

$$= r^4 k^4 \left(\frac{1}{r^2 k^4} - \frac{1}{r^2 k^2} + \frac{1}{r^2} \right)$$

$$[\because q = rk \text{ and } p = rk^2]$$

$$= r^2 - r^2 k^2 + r^2 k^4$$

$$= r^2 - (rk)^2 + (rk^2)^2$$

$$= r^2 - q^2 + p^2$$

$$[\because q = rk \text{ and } p = rk^2]$$

$$= \text{L.H.S.}$$

Hence Proved.

21 If $\frac{a}{b} = \frac{c}{d}$, prove that each given ratio ($\frac{a}{b}$ and $\frac{c}{d}$) is equal to :

(i) $\frac{3a-5c}{3b-5d}$

(ii) $\sqrt{\frac{2a^2+9c^2}{2b^2+9d^2}}$

(iii) $\left(\frac{5a^3-13c^3}{5b^3-13d^3}\right)^{\frac{1}{3}}$

Solution :

$$\text{Let } \frac{a}{b} = \frac{c}{d} = k \Rightarrow \frac{a}{b} = k \text{ and } \frac{c}{d} = k \Rightarrow a = bk \text{ and } c = dk$$

(i) $\frac{3a-5c}{3b-5d} = \frac{3(bk)-5(dk)}{3b-5d} = \frac{k(3b-5d)}{3b-5d} = k = \text{each given ratio}$

Hence Proved.

(ii) $\sqrt{\frac{2a^2+9c^2}{2b^2+9d^2}} = \sqrt{\frac{2(bk)^2+9(dk)^2}{2b^2+9d^2}} = \sqrt{\frac{k^2(2b^2+9d^2)}{2b^2+9d^2}}$
 $= \sqrt{k^2} = k = \text{each given ratio}$

Hence Proved.

(iii) $\left(\frac{5a^3-13c^3}{5b^3-13d^3}\right)^{\frac{1}{3}} = \left[\frac{5(bk)^3-13(dk)^3}{5b^3-13d^3}\right]^{\frac{1}{3}} = \left[\frac{k^3(5b^3-13d^3)}{5b^3-13d^3}\right]^{\frac{1}{3}}$
 $= [k^3]^{\frac{1}{3}} = k = \text{each given ratio}$

Hence Proved.

22 If a, b, c and d are in proportion, prove that :

(i) $\frac{a-b}{c-d} = \sqrt{\frac{3a^2+8b^2}{3c^2+8d^2}}$

(ii) $\frac{5a^2+12c^2}{5b^2+12d^2} = \sqrt{\frac{3a^4-7c^4}{3b^4-7d^4}}$

Solution :

a, b, c and d are in proportion

$$\Rightarrow \frac{a}{b} = \frac{c}{d} = k \text{ (let)}$$

$$\Rightarrow \frac{a}{b} = k \text{ and } \frac{c}{d} = k \quad \text{i.e. } a = bk \text{ and } c = dk$$

(i) L.H.S. $= \frac{a-b}{c-d} = \frac{bk-b}{dk-d} = \frac{b(k-1)}{d(k-1)} = \frac{b}{d} \quad \dots \text{I}$

R.H.S. $= \sqrt{\frac{3a^2+8b^2}{3c^2+8d^2}} = \sqrt{\frac{3(bk)^2+8b^2}{3(dk)^2+8d^2}}$
 $= \sqrt{\frac{b^2(3k^2+8)}{d^2(3k^2+8)}} = \sqrt{\frac{b^2}{d^2}} = \frac{b}{d} \quad \dots \text{II}$

From equations I and II, we get : **L.H.S. = R.H.S.**

Hence Proved.

$$\begin{aligned}
 \text{(ii)} \quad \text{L.H.S.} &= \frac{5a^2 + 12c^2}{5b^2 + 12d^2} = \frac{5(bk)^2 + 12(dk)^2}{5b^2 + 12d^2} \\
 &= \frac{k^2(5b^2 + 12d^2)}{5b^2 + 12d^2} = k^2 \quad \dots \text{I} \\
 \text{R.H.S.} &= \sqrt{\frac{3a^4 - 7c^4}{3b^4 - 7d^4}} = \sqrt{\frac{3(bk)^4 - 7(dk)^4}{3b^4 - 7d^4}} \\
 &= \sqrt{\frac{k^4(3b^4 - 7d^4)}{3b^4 - 7d^4}} = \sqrt{k^4} = k^2 \quad \dots \text{II}
 \end{aligned}$$

From equations I and II, we get : **L.H.S. = R.H.S.**

Hence Proved.

- 23** 6 is the mean proportion between two numbers x and y and 48 is third proportion to x and y . Find the numbers. [2011]

Solution :

Since, 6 is mean proportional between x and y .

$$\Rightarrow x : 6 = 6 : y \quad \Rightarrow xy = 36 \quad \dots \text{I}$$

and, 48 is third proportional to x and y

$$\Rightarrow x : y = y : 48 \quad \Rightarrow y^2 = 48x \quad \dots \text{II}$$

$$\text{From eq. (I); } xy = 36 \quad \Rightarrow x = \frac{36}{y}$$

Substituting $x = \frac{36}{y}$ in eq. II, we get :

$$y^2 = 48 \times \frac{36}{y} \quad \Rightarrow y^3 = 36 \times 48 \text{ and, } y = 12$$

$$\therefore x = \frac{36}{y} = \frac{36}{12} = 3$$

$\therefore$ The required nos. are 3 and 12.

Ans.

EXERCISE 7(B)

- Find the fourth proportional to :
 - 1.5, 4.5 and 3.5
 - $3a$, $6a^2$ and $2ab^2$
- Find the third proportional to :
 - $2\frac{2}{3}$ and 4
 - $a - b$ and $a^2 - b^2$
- Find the mean proportional between :
 - $6 + 3\sqrt{3}$ and $8 - 4\sqrt{3}$
 - $a - b$ and $a^3 - a^2b$
- If $x + 5$ is the mean proportion between $x + 2$ and $x + 9$; find the value of x .
- If x^2 , 4 and 9 are in continued proportion, find x .
- What least number must be added to each of the numbers 6, 15, 20 and 43 to make them proportional ? (2005, 2013)
- (i) If a , b , c are in continued proportion, show that : $\frac{a^2 + b^2}{b(a+c)} = \frac{b(a+c)}{b^2 + c^2}$.

- (ii) If a, b, c are in continued proportion and $a(b - c) = 2b$, prove that :

$$a - c = \frac{2(a+b)}{a}$$

- (iii) If $\frac{a}{b} = \frac{c}{d}$, show that :

$$\frac{a^3c + ac^3}{b^3d + bd^3} = \frac{(a+c)^4}{(b+d)^4}$$

8. What least number must be subtracted from each of the numbers 7, 17 and 47 so that the remainders are in continued proportion ?
9. If y is the mean proportional between x and z ; show that $xy + yz$ is the mean proportional between $x^2 + y^2$ and $y^2 + z^2$.
10. If q is the mean proportional between p and r , show that :
- $$pqr(p + q + r)^3 = (pq + qr + pr)^3$$

q is the mean proportional between p and $r \Rightarrow q^2 = pr$.

$$\begin{aligned} \text{L.H.S.} &= pqr(p + q + r)^3 \\ &= q \cdot q^2(p + q + r)^3 \quad \because pr = q^2 \\ &= q^3(p + q + r)^3 \\ &= [q(p + q + r)]^3 \\ &= (pq + q^2 + qr)^3 \\ &= (pq + pr + qr)^3 = \text{R.H.S.} \end{aligned}$$

11. If three quantities are in continued proportion; show that the ratio of the first to the third is the duplicate ratio of the first to the second.

If the three quantities be x, y and z ; then $x : y = y : z$ and to prove that $x : z = x^2 : y^2$

12. If y is the mean proportional between x and z , prove that : $\frac{x^2 - y^2 + z^2}{x^{-2} - y^{-2} + z^{-2}} = y^4$.
13. Given four quantities a, b, c and d are in proportion. Show that :

$$\begin{aligned} (a - c) b^2 : (b - d) cd \\ = (a^2 - b^2 - ab) : (c^2 - d^2 - cd) \end{aligned}$$

$$\text{Given : } \frac{a}{b} = \frac{c}{d} = k \text{ (let)}$$

$$\Rightarrow a = bk \text{ and } c = dk$$

Now, find the values of L.H.S. and R.H.S. of the required result by substituting $a = bk$ and $c = dk$; and show **L.H.S. = R.H.S.**

14. Find two numbers such that the mean proportional between them is 12 and the third proportional to them is 96.
15. Find the third proportional to $\frac{x}{y} + \frac{y}{x}$ and $\sqrt{x^2 + y^2}$
16. If $p : q = r : s$; then show that :
- $$mp + nq : q = mr + ns : s$$

$$\begin{aligned} \frac{p}{q} = \frac{r}{s} &\Rightarrow \frac{mp}{q} = \frac{mr}{s} \\ \Rightarrow \frac{mp}{q} + n &= \frac{mr}{s} + n \text{ and so on.} \end{aligned}$$

17. If $p + r = mq$ and $\frac{1}{q} + \frac{1}{s} = \frac{m}{r}$; then prove that : $p : q = r : s$.
18. If $\frac{a}{b} = \frac{c}{d}$, prove that each of the given ratio is equal to :

$$(i) \frac{5a + 4c}{5b + 4d}$$

$$(ii) \frac{13a - 8c}{13b - 8d}$$

$$(iii) \sqrt{\frac{3a^2 - 10c^2}{3b^2 - 10d^2}}$$

$$(iv) \left(\frac{8a^3 + 15c^3}{8b^3 + 15d^3} \right)^{\frac{1}{3}}$$

19. If a, b, c and d are in proportion, prove that :

$$(i) \frac{13a + 17b}{13c + 17d} = \sqrt{\frac{2ma^2 - 3nb^2}{2mc^2 - 3nd^2}}$$

$$(ii) \sqrt{\frac{4a^2 + 9b^2}{4c^2 + 9d^2}} = \left(\frac{xa^3 - 5yb^3}{xc^3 - 5yd^3} \right)^{\frac{1}{3}}$$

20. If $\frac{x}{a} = \frac{y}{b} = \frac{z}{c}$, prove that :

$$\frac{2x^3 - 3y^3 + 4z^3}{2a^3 - 3b^3 + 4c^3} = \left(\frac{2x - 3y + 4z}{2a - 3b + 4c} \right)^3$$

7.8 Some Important Properties of Proportion :

If four quantities a, b, c and d form a proportion

i.e. if $a : b :: c : d$, many other proportions may be obtained using the properties of fractions. Some of these proportions are given below :

1. Invertendo :

According to this property of proportions :

If $a : b = c : d$, then $b : a = d : c$.

$$\begin{aligned}
 \text{Proof : } a : b = c : d &\Rightarrow \frac{a}{b} = \frac{c}{d} \\
 &\Rightarrow \frac{b}{a} = \frac{d}{c} && [\text{Taking reciprocal of both the sides}] \\
 &\Rightarrow b : a = d : c.
 \end{aligned}$$

2. Alternendo :

According to this property of proportions :

If $a : b = c : d$, then $a : c = b : d$.

$$\begin{aligned}
 \text{Proof : } a : b = c : d &\Rightarrow \frac{a}{b} = \frac{c}{d} \\
 &\Rightarrow a \times d = b \times c && [\text{By cross-multiplication}] \\
 &\Rightarrow \frac{a}{c} = \frac{b}{d} \Rightarrow a : c = b : d.
 \end{aligned}$$

3. Componendo :

If $a : b = c : d$, then $a + b : b = c + d : d$.

$$\begin{aligned}
 \text{Proof : } a : b = c : d &\Rightarrow \frac{a}{b} = \frac{c}{d} \\
 &\Rightarrow \frac{a}{b} + 1 = \frac{c}{d} + 1 && [\text{Adding 1 on each side}] \\
 &\Rightarrow \frac{a+b}{b} = \frac{c+d}{d} \\
 &\Rightarrow a + b : b = c + d : d
 \end{aligned}$$

4. Dividendo :

If $a : b = c : d$, then $a - b : b = c - d : d$.

$$\begin{aligned}
 \text{Proof : } a : b = c : d &\Rightarrow \frac{a}{b} = \frac{c}{d} \\
 &\Rightarrow \frac{a}{b} - 1 = \frac{c}{d} - 1 && [\text{Subtracting 1 from each side}] \\
 &\Rightarrow \frac{a-b}{b} = \frac{c-d}{d} \\
 &\Rightarrow a - b : b = c - d : d.
 \end{aligned}$$

5. Componendo and Dividendo :

If $a : b = c : d$, then $a + b : a - b = c + d : c - d$.

By componendo, $a : b = c : d \Rightarrow a + b : b = c + d : d$

$$\Rightarrow \frac{a+b}{b} = \frac{c+d}{d} \quad \dots\dots I$$

By dividendo, $a : b = c : d \Rightarrow \frac{a-b}{b} = \frac{c-d}{d}$ II

Dividing I by II, we get : $\frac{a+b}{a-b} = \frac{c+d}{c-d}$

Thus, $a : b = c : d \Rightarrow a + b : a - b = c + d : c - d$.

Thus; $\frac{a}{b} = \frac{c}{d} \Rightarrow$ (i) $\frac{b}{a} = \frac{d}{c}$ By Invertendo

(ii) $\frac{a}{c} = \frac{b}{d}$ By Alternendo

(iii) $\frac{a+b}{b} = \frac{c+d}{d}$ By Componendo

(iv) $\frac{a-b}{b} = \frac{c-d}{d}$ By Dividendo

(v) $\frac{a+b}{a-b} = \frac{c+d}{c-d}$ By Componendo and Dividendo

7.9 Direct Applications :

24 If $\frac{8x+13y}{8x-13y} = \frac{9}{7}$, find $x : y$.

Solution :

Applying componendo and dividendo :

$$\frac{8x+13y}{8x-13y} = \frac{9}{7} \text{ gives } \frac{8x+13y+8x-13y}{8x+13y-8x+13y} = \frac{9+7}{9-7}$$

i.e. $\frac{16x}{26y} = \frac{16}{2} \Rightarrow \frac{x}{y} = \frac{16}{2} \times \frac{26}{16} = \frac{13}{1}$ i.e. $x : y = 13 : 1$ **Ans.**

Alternative method :

$$\frac{8x+13y}{8x-13y} = \frac{9}{7} \Rightarrow 72x - 117y = 56x + 91y$$

$$\Rightarrow 16x = 208y$$

$$\Rightarrow \frac{x}{y} = \frac{208}{16} = \frac{13}{1} \text{ i.e. } x : y = 13 : 1 \text{ **Ans.**}$$

25 If $a : b = c : d$, show that : $3a + 2b : 3a - 2b = 3c + 2d : 3c - 2d$.

Solution :

$$a : b = c : d \Rightarrow \frac{a}{b} = \frac{c}{d}$$

$$\Rightarrow \frac{3a}{2b} = \frac{3c}{2d} \quad [\text{Multiplying each side by } \frac{3}{2}]$$

$$\Rightarrow \frac{3a+2b}{3a-2b} = \frac{3c+2d}{3c-2d} \quad [\text{By componendo and dividendo}]$$

$$\Rightarrow 3a+2b : 3a-2b = 3c+2d : 3c-2d \quad \text{Ans.}$$

Alternative method :

$$a : b = c : d \Rightarrow \frac{a}{b} = \frac{c}{d} = k \text{ (let)}$$

$$\Rightarrow a = bk \text{ and } c = dk$$

$$\therefore 3a+2b : 3a-2b = \frac{3a+2b}{3a-2b} = \frac{3bk+2b}{3bk-2b} \quad [\text{As } a = bk]$$

$$= \frac{b(3k+2)}{b(3k-2)} = \frac{3k+2}{3k-2} \quad \dots \text{I}$$

$$\text{and } 3c+2d : 3c-2d = \frac{3c+2d}{3c-2d} = \frac{3dk+2d}{3dk-2d} \quad [\text{As } c = dk]$$

$$= \frac{d(3k+2)}{d(3k-2)} = \frac{3k+2}{3k-2} \quad \dots \text{II}$$

From I and II, we get :

$$3a+2b : 3a-2b = 3c+2d : 3c-2d \quad \text{Ans.}$$

26 If $\frac{8a-5b}{8c-5d} = \frac{8a+5b}{8c+5d}$, prove that $\frac{a}{b} = \frac{c}{d}$. [2008]

Solution :

$$\text{Given : } \frac{8a-5b}{8c-5d} = \frac{8a+5b}{8c+5d}$$

$$\Rightarrow \frac{8a-5b}{8a+5b} = \frac{8c-5d}{8c+5d} \quad [\text{Applying alternendo}]$$

$$\Rightarrow \frac{8a-5b+8a+5b}{8a-5b-8a-5b} = \frac{8c-5d+8c+5d}{8c-5d-8c-5d} \quad [\text{Applying componendo and dividendo}]$$

$$\Rightarrow \frac{16a}{-10b} = \frac{16c}{-10d}$$

$$\Rightarrow \frac{a}{b} = \frac{c}{d}$$

Hence Proved.

27 If $p = \frac{4xy}{x+y}$, find the value of $\frac{p+2x}{p-2x} + \frac{p+2y}{p-2y}$.

Solution :

$$p = \frac{4xy}{x+y} \Rightarrow \frac{P}{2x} = \frac{2y}{x+y} \quad [\text{Now apply componendo and dividendo}]$$

$$\Rightarrow \frac{p+2x}{p-2x} = \frac{2y+x+y}{2y-x-y} = \frac{x+3y}{y-x}$$

$$\text{Again, } p = \frac{4xy}{x+y} \Rightarrow \frac{P}{2y} = \frac{2x}{x+y} \quad [\text{Now apply componendo and dividendo}]$$

$$\Rightarrow \frac{p+2y}{p-2y} = \frac{2x+x+y}{2x-x-y} = \frac{3x+y}{x-y}$$

$$\therefore \frac{p+2x}{p-2x} + \frac{p+2y}{p-2y} = \frac{x+3y}{y-x} + \frac{3x+y}{x-y}$$

$$= \frac{x+3y}{y-x} - \frac{3x+y}{y-x} = \frac{x+3y-3x-y}{y-x} = 2 \quad \text{Ans.}$$

Alternative method :

$$\frac{p+2x}{p-2x} + \frac{p+2y}{p-2y} = \frac{\frac{4xy}{x+y} + 2x}{\frac{4xy}{x+y} - 2x} + \frac{\frac{4xy}{x+y} + 2y}{\frac{4xy}{x+y} - 2y}$$

$$= \frac{4xy+2x(x+y)}{4xy-2x(x+y)} + \frac{4xy+2y(x+y)}{4xy-2y(x+y)}$$

$$= \frac{4xy+2x^2+2xy}{4xy-2x^2-2xy} + \frac{4xy+2xy+2y^2}{4xy-2xy-2y^2}$$

$$= \frac{6xy+2x^2}{2xy-2x^2} + \frac{6xy+2y^2}{2xy-2y^2}$$

$$= \frac{2x(3y+x)}{2x(y-x)} + \frac{2y(3x+y)}{2y(x-y)}$$

$$= \frac{3y+x}{y-x} - \frac{3x+y}{y-x} \quad \left[\because \frac{3x+y}{x-y} = -\frac{3x+y}{y-x} \right]$$

$$= \frac{3y+x-3x-y}{y-x} = \frac{2y-2x}{y-x} = \frac{2(y-x)}{y-x} = 2 \quad \text{Ans.}$$

28 If $a : b = c : d$; prove that :

$$(a^2 + ac + c^2) : (a^2 - ac + c^2) = (b^2 + bd + d^2) : (b^2 - bd + d^2)$$

Solution :

$$a : b = c : d \Rightarrow \frac{a}{b} = \frac{c}{d} = k \text{ (say)} \Rightarrow a = bk \text{ and } c = dk$$

$$\therefore (a^2 + ac + c^2) : (a^2 - ac + c^2) = \frac{a^2 + ac + c^2}{a^2 - ac + c^2}$$

$$\begin{aligned}
 &= \frac{b^2 k^2 + (bk)(dk) + d^2 k^2}{b^2 k^2 - (bk)(dk) + d^2 k^2} \quad [\because a = bk \text{ and } c = dk] \\
 &= \frac{k^2 (b^2 + bd + d^2)}{k^2 (b^2 - bd + d^2)} \\
 &= (b^2 + bd + d^2) : (b^2 - bd + d^2)
 \end{aligned}$$

Hence Proved.

29 If x , y and z are in continued proportion, prove that :

$$x^2 - y^2 : x^2 + y^2 = x - z : x + z.$$

Solution :

x , y and z are in continued proportion

$$\Rightarrow \frac{x}{y} = \frac{y}{z} = k \text{ (say)} \Rightarrow x = yk, y = zk \text{ and } x = yk = (zk)k = zk^2$$

$$\begin{aligned}
 \therefore x^2 - y^2 : x^2 + y^2 &= \frac{x^2 - y^2}{x^2 + y^2} = \frac{y^2 k^2 - y^2}{y^2 k^2 + y^2} \quad [\because x = yk] \\
 &= \frac{y^2 (k^2 - 1)}{y^2 (k^2 + 1)} = \frac{k^2 - 1}{k^2 + 1} \quad \dots \text{I}
 \end{aligned}$$

$$\begin{aligned}
 \text{Also, } x - z : x + z &= \frac{x - z}{x + z} = \frac{zk^2 - z}{zk^2 + z} \quad [\because x = zk^2] \\
 &= \frac{z (k^2 - 1)}{z (k^2 + 1)} = \frac{k^2 - 1}{k^2 + 1} \quad \dots \text{II}
 \end{aligned}$$

From I and II, we get :

$$x^2 - y^2 : x^2 + y^2 = x - z : x + z$$

Hence Proved.

Alternative method :

$$x, y \text{ and } z \text{ are in continued proportion} \Rightarrow \frac{x}{y} = \frac{y}{z} \Rightarrow y^2 = xz.$$

$$\therefore x^2 - y^2 : x^2 + y^2 = \frac{x^2 - y^2}{x^2 + y^2} = \frac{x^2 - xz}{x^2 + xz} \quad [\because y^2 = xz]$$

$$= \frac{x(x - z)}{x(x + z)} = \frac{x - z}{x + z} = x - z : x + z \quad \text{Hence Proved.}$$

30 Using the properties of proportion, solve the following equation for x :

$$\frac{x^3 + 3x}{3x^2 + 1} = \frac{341}{91}$$

Solution :

Applying componendo and dividendo, we get :

$$\frac{x^3 + 3x + 3x^2 + 1}{x^3 + 3x - 3x^2 - 1} = \frac{341 + 91}{341 - 91}$$

$$\Rightarrow \frac{(x+1)^3}{(x-1)^3} = \frac{432}{250} = \frac{216}{125} = \left(\frac{6}{5}\right)^3$$

$$\Rightarrow \frac{x+1}{x-1} = \frac{6}{5}$$

Again, applying componendo and dividendo, we get :

$$\frac{x+1+x-1}{x+1-x+1} = \frac{6+5}{6-5} \quad \text{i.e.,} \quad \frac{2x}{2} = \frac{11}{1} \Rightarrow x = 11$$

Ans.

31 If $x = \frac{\sqrt{3a+2b} + \sqrt{3a-2b}}{\sqrt{3a+2b} - \sqrt{3a-2b}}$, prove that : $bx^2 - 3ax + b = 0$

Solution :

Given : $\frac{x}{1} = \frac{\sqrt{3a+2b} + \sqrt{3a-2b}}{\sqrt{3a+2b} - \sqrt{3a-2b}}$

$$\Rightarrow \frac{x+1}{x-1} = \frac{\sqrt{3a+2b} + \sqrt{3a-2b} + \sqrt{3a+2b} - \sqrt{3a-2b}}{\sqrt{3a+2b} + \sqrt{3a-2b} - \sqrt{3a+2b} + \sqrt{3a-2b}}$$

[Applying componendo and dividendo]

$$\Rightarrow \frac{x+1}{x-1} = \frac{2\sqrt{3a+2b}}{2\sqrt{3a-2b}}$$

$$\Rightarrow \frac{x+1}{x-1} = \frac{\sqrt{3a+2b}}{\sqrt{3a-2b}}$$

$$\Rightarrow \frac{x^2+2x+1}{x^2-2x+1} = \frac{3a+2b}{3a-2b} \quad \text{[Squaring both the sides]}$$

$$\Rightarrow \frac{x^2+2x+1+x^2-2x+1}{x^2+2x+1-x^2+2x-1} = \frac{3a+2b+3a-2b}{3a+2b-3a+2b} \quad \text{[By componendo and dividendo]}$$

$$\Rightarrow \frac{2(x^2+1)}{4x} = \frac{6a}{4b}$$

$$\Rightarrow \frac{x^2+1}{2x} = \frac{3a}{2b} \quad \text{i.e.,} \quad 2bx^2 + 2b = 6ax$$

$$\Rightarrow bx^2 - 3ax + b = 0$$

Hence Proved.

Alternative method :

Given : $x = \frac{\sqrt{3a+2b} + \sqrt{3a-2b}}{\sqrt{3a+2b} - \sqrt{3a-2b}}$

$$\Rightarrow x = \frac{\sqrt{3a+2b} + \sqrt{3a-2b}}{\sqrt{3a+2b} - \sqrt{3a-2b}} \times \frac{\sqrt{3a+2b} + \sqrt{3a-2b}}{\sqrt{3a+2b} + \sqrt{3a-2b}}$$

$$= \frac{3a+2b+3a-2b+2\sqrt{(3a+2b)(3a-2b)}}{3a+2b-3a+2b}$$

$$= \frac{6a+2\sqrt{9a^2-4b^2}}{4b} = \frac{3a+\sqrt{9a^2-4b^2}}{2b}$$

$$\Rightarrow 2bx = 3a + \sqrt{9a^2 - 4b^2}$$

$$\Rightarrow 2bx - 3a = \sqrt{9a^2 - 4b^2}$$

$$\Rightarrow 4b^2x^2 + 9a^2 - 12abx = 9a^2 - 4b^2$$

[Squaring]

$$\Rightarrow 4b^2x^2 - 12abx + 4b^2 = 0 \Rightarrow bx^2 - 3ax + b = 0$$

Hence Proved.

EXERCISE 7(C)1. If $a : b = c : d$, prove that :

$$(i) 5a + 7b : 5a - 7b = 5c + 7d : 5c - 7d.$$

$$(ii) (9a + 13b) (9c - 13d) \\ = (9c + 13d) (9a - 13b).$$

$$(iii) xa + yb : xc + yd = b : d$$

2. If $a : b = c : d$, prove that :

$$(6a + 7b) (3c - 4d) = (6c + 7d) (3a - 4b).$$

3. Given, $\frac{a}{b} = \frac{c}{d}$, prove that :

$$\frac{3a - 5b}{3a + 5b} = \frac{3c - 5d}{3c + 5d}$$

[2000]

$$4. \text{ If } \frac{5x + 6y}{5u + 6v} = \frac{5x - 6y}{5u - 6v};$$

then prove that $x : y = u : v$.

$$5. \text{ If } (7a + 8b) (7c - 8d)$$

$$= (7a - 8b) (7c + 8d);$$

prove that $a : b = c : d$.

$$6. (i) \text{ If } x = \frac{6ab}{a + b}, \text{ find the value of :}$$

$$\frac{x + 3a}{x - 3a} + \frac{x + 3b}{x - 3b}.$$

$$(ii) \text{ If } a = \frac{4\sqrt{6}}{\sqrt{2} + \sqrt{3}}, \text{ find the value of :}$$

$$\frac{a + 2\sqrt{2}}{a - 2\sqrt{2}} + \frac{a + 2\sqrt{3}}{a - 2\sqrt{3}}.$$

$$7. \text{ If } (a + b + c + d) (a - b - c + d)$$

$$= (a + b - c - d) (a - b + c - d);$$

prove that : $a : b = c : d$.

$$8. \text{ If } \frac{a - 2b - 3c + 4d}{a + 2b - 3c - 4d}$$

$$= \frac{a - 2b + 3c - 4d}{a + 2b + 3c + 4d},$$

show that : $2ad = 3bc$.

$$9. \text{ If } (a^2 + b^2) (x^2 + y^2) = (ax + by)^2;$$

$$\text{prove that : } \frac{a}{x} = \frac{b}{y}.$$

10. If a, b and c are in continued proportion, prove that :

$$(i) \frac{a^2 + ab + b^2}{b^2 + bc + c^2} = \frac{a}{c}$$

$$(ii) \frac{a^2 + b^2 + c^2}{(a + b + c)^2} = \frac{a - b + c}{a + b + c}.$$

11. Using properties of proportion, solve for x :

$$(i) \frac{\sqrt{x+5} + \sqrt{x-16}}{\sqrt{x+5} - \sqrt{x-16}} = \frac{7}{3}.$$

$$(ii) \frac{\sqrt{x+1} + \sqrt{x-1}}{\sqrt{x+1} - \sqrt{x-1}} = \frac{4x-1}{2}.$$

$$(iii) \frac{3x + \sqrt{9x^2 - 5}}{3x - \sqrt{9x^2 - 5}} = 5.$$

$$12. \text{ If } x = \frac{\sqrt{a+3b} + \sqrt{a-3b}}{\sqrt{a+3b} - \sqrt{a-3b}}, \text{ prove that :}$$

$$3bx^2 - 2ax + 3b = 0.$$

[2007]

13. Using the properties of proportion, solve for

$$x, \text{ given } \frac{x^4 + 1}{2x^2} = \frac{17}{8}.$$

[2013]

14. If $x = \frac{\sqrt{m+n} + \sqrt{m-n}}{\sqrt{m+n} - \sqrt{m-n}}$, express n in terms of x and m .

15. If $\frac{x^3 + 3xy^2}{3x^2y + y^3} = \frac{m^3 + 3mn^2}{3m^2n + n^3}$,
show that : $nx = my$.

EXERCISE 7(D)

- If $a : b = 3 : 5$, find :
(10a + 3b) : (5a + 2b)
- If $5x + 6y : 8x + 5y = 8 : 9$, find : $x : y$.
- If $(3x - 4y) : (2x - 3y) = (5x - 6y) : (4x - 5y)$, find : $x : y$.
- Find the :
(i) duplicate ratio of $2\sqrt{2} : 3\sqrt{5}$
(ii) triplicate ratio of $2a : 3b$,
(iii) sub-duplicate ratio of $9x^2a^4 : 25y^6b^2$
(iv) sub-triplicate ratio of $216 : 343$
(v) reciprocal ratio of $3 : 5$
(vi) ratio compounded of the duplicate ratio of $5 : 6$, the reciprocal ratio of $25 : 42$ and the sub-duplicate ratio of $36 : 49$.
- Find the value of x , if :
(i) $(2x + 3) : (5x - 38)$ is the duplicate ratio of $\sqrt{5} : \sqrt{6}$.
(ii) $(2x + 1) : (3x + 13)$ is the sub-duplicate ratio of $9 : 25$.
(iii) $(3x - 7) : (4x + 3)$ is the sub-triplicate ratio of $8 : 27$.
- What quantity must be added to each term of the ratio $x : y$ so that it may become equal to $c : d$?
- A woman reduces her weight in the ratio $7 : 5$. What does her weight become if originally it was 84 kg?
- If $15(2x^2 - y^2) = 7xy$, find $x : y$; if x and y both are positive.
- Find the :
(i) fourth proportional to $2xy$, x^2 and y^2 .
(ii) third proportional to $a^2 - b^2$ and $a + b$.
(iii) mean proportion to $(x - y)$ and $(x^3 - x^2y)$
- Find two numbers such that the mean proportional between them is 14 and third proportional to them is 112.
- If x and y be unequal and $x : y$ is the duplicate ratio of $x + z$ and $y + z$, prove that z is mean proportional between x and y .
- If q is the mean proportional between p and r , prove that : $\frac{p^3 + q^3 + r^3}{p^2q^2r^2} = \frac{1}{p^3} + \frac{1}{q^3} + \frac{1}{r^3}$.
- If a , b and c are in continued proportion, prove that : $a : c = (a^2 + b^2) : (b^2 + c^2)$.
- If $x = \frac{2ab}{a+b}$, find the value of : $\frac{x+a}{x-a} + \frac{x+b}{x-b}$.
- If $(4a + 9b)(4c - 9d) = (4a - 9b)(4c + 9d)$, prove that : $a : b = c : d$.
- If $\frac{a}{b} = \frac{c}{d}$, show that :
 $(a + b) : (c + d) = \sqrt{a^2 + b^2} : \sqrt{c^2 + d^2}$
- If $\frac{x}{a} = \frac{y}{b} = \frac{z}{c}$, prove that :
 $\frac{ax - by}{(a+b)(x-y)} + \frac{by - cz}{(b+c)(y-z)} + \frac{cz - ax}{(c+a)(z-x)} = 3$
- There are 36 members in a student council in a school and the ratio of the number of boys to the number of girls is $3 : 1$. How many more girls should be added to the council so that the ratio of number of boys to the number of girls may be $9 : 5$?
- If $7x - 15y = 4x + y$, find the value of $x : y$. Hence, use componendo and dividendo to find the values of :
(i) $\frac{9x + 5y}{9x - 5y}$ (ii) $\frac{3x^2 + 2y^2}{3x^2 - 2y^2}$
- If $\frac{4m+3n}{4m-3n} = \frac{7}{4}$, use properties of proportion to find :
(i) $m : n$ (ii) $\frac{2m^2 - 11n^2}{2m^2 + 11n^2}$.

21. If x, y, z are in continued proportion, prove

that : $\frac{(x+y)^2}{(y+z)^2} = \frac{x}{z}$. [2010]

22. Given $x = \frac{\sqrt{a^2+b^2} + \sqrt{a^2-b^2}}{\sqrt{a^2+b^2} - \sqrt{a^2-b^2}}$.

Use componendo and dividendo to prove that :

$b^2 = \frac{2a^2x}{x^2+1}$. [2010]

23. If $\frac{x^2+y^2}{x^2-y^2} = 2\frac{1}{8}$, find :

(i) $\frac{x}{y}$ (ii) $\frac{x^3+y^3}{x^3-y^3}$ [2014]

24. Using componendo and dividendo, find the value of x :

$\frac{\sqrt{3x+4} + \sqrt{3x-5}}{\sqrt{3x+4} - \sqrt{3x-5}} = 9$. [2011]

25. If $x = \frac{\sqrt{a+1} + \sqrt{a-1}}{\sqrt{a+1} - \sqrt{a-1}}$, using properties of proportion show that :

$x^2 - 2ax + 1 = 0$. [2012]

26. Given $\frac{x^3+12x}{6x^2+8} = \frac{y^3+27y}{9y^2+27}$. Using componendo and dividendo, find $x : y$.

[2015]